

CATHACTION: Endovascular Intervention Tool Segmentation and Collision Detection: Structured description of the challenge design

CHALLENGE ORGANIZATION

Title

Use the title to convey the essential information on the challenge mission.

CATHACTION: Endovascular Intervention Tool Segmentation and Collision Detection

Challenge acronym

Preferable, provide a short acronym of the challenge (if any).

CATHACTION

Challenge abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

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Cardiovascular disease remains one of the leading causes of mortality worldwide. Minimally invasive endovascular interventions have emerged as the standard of care due to reduced trauma, lower complication rates, and faster post-operative recovery compared to traditional open surgery. However, navigating catheters and guidewires through the vasculature demands highly specialized skills; unintended collisions with vessel walls can result in perforation, hemorrhage, or life-threatening complications. Although recent advances in surgical AI have shown promise, progress in endovascular assistance is limited by the absence of large, diverse, and clinically realistic datasets for benchmarking algorithmic performance.

To address this need, we present CATHACTION, a new multi-task MICCAI challenge designed to advance surgical perception and intraoperative support for endovascular interventions. CATHACTION provides over 600,000 frames, making it one of the largest datasets in this domain. Critically, the data come from three distinct collection environments: (1) silicon phantom training systems, (2) animal X-ray fluoroscopy, and (3) real human X-ray procedures—yielding rich variability in image appearance, anatomy, and instrumentation. This unique multi-domain coverage enables the study of domain generalization, domain adaptation, and transfer learning, all of which are essential for deploying ML systems in real clinical settings.

The 2026 MICCAI CATHACTION Challenge will focus on two core tasks: (i) Tool Segmentation, and (ii) Collision Detection. Participants are encouraged to exploit the multi-domain nature and scale of the dataset, and to investigate models capable of cross-domain generalization, joint learning, or foundation-style representations.

Challenge keywords

List the primary keywords that characterize the challenge.

Endovascular Intervention, Catheter Segmentation, Collision Detection

Year

2026

Novelty of the challenge

Briefly describe the novelty of the challenge.

The CATHACTION challenge provides a fundamentally new benchmark for research in endovascular surgical AI by filling a conspicuous gap in the landscape of surgical datasets and challenges. Although surgical vision tasks for endoscopic and laparoscopic procedures have benefited greatly from standardized benchmarks, the field of fluoroscopy-guided endovascular intervention remains largely underserved, despite its high clinical volume, technical complexity, and sensitivity to procedural errors. Existing datasets for catheter and guidewire analysis are typically small, single-domain, and task-specific, which prevents rigorous evaluation of model robustness, cross-domain transfer, and clinical deployment feasibility.

In contrast, CATHACTION introduces a large-scale dataset exceeding 600,000 fluoroscopic frames and critically, aggregates data from three separate and independently acquired domains: silicon phantom systems, animal X-ray studies, and real human X-ray procedures. This design enables systematic investigation of the domain adaptation and generalization problem across multiple axes of shift, including synthetic-to-real, cross-species physiological differences, and intra-clinical variations in imaging characteristics, anatomy, and instrumentation. Such shifts are known to cause significant performance degradation for machine learning models, and addressing them is essential for achieving trustworthy real-world deployment in safety-critical applications.

Beyond its data diversity, CATHACTION is the first benchmark to jointly evaluate catheter/guidewire segmentation and collision detection within the same challenge framework. These tasks are not only clinically meaningful and directly tied to procedural safety, but also interdependent and multi-nodal from an AI perspective, requiring perception systems that can simultaneously localize tools, reason about spatial context, and predict potential hazardous interactions. By pairing a multi-domain dataset (i.e., data from silicon phantom, animal, and real-human) with clinically relevant tasks and standardized evaluation protocols, CATHACTION provides a unique testbed for multi-task learning, scalable representation learning, and foundation-model style approaches. In doing so, it elevates the research agenda from narrow task performance toward robust and generalizable vision systems that can transfer across domains and ultimately support real clinical workflows.

Task description and application scenarios

Briefly describe the application scenarios for the tasks in the challenge.

N/A

FURTHER INFORMATION FOR CONFERENCE ORGANIZERS

Workshop

If the challenge is part of a workshop, please indicate the workshop.

N/A

Duration

How long does the challenge take?

2 Hours

In case you selected half or full day, please explain why you need a long slot for your challenge.

N/A

Expected number of participants

Please explain the basis of your estimate (e.g. numbers from previous challenges) and/or provide a list of potential participants and indicate if they have already confirmed their willingness to contribute.

The CATHACTION challenge is expected to attract contributions from academic and industrial researchers in medical imaging, computer vision, and medical robotics. The scope and structure of the challenge are based on insights gained from previous large-scale medical imaging challenges, such as those organized by MICCAI. We estimate that approximately 60 participants will engage in the two tasks.

Publication and future plans

Please indicate if you plan to coordinate a publication of the challenge results.

We plan to publish the joint results of the CATHACTION challenge in a top journal in medical imaging within 8 months of the challenge. Participating teams will have the opportunity to publish their methods separately, but only after the joint journal publication. The embargo period for individual publications will be 10 months.

MICCAI LNCS proceedings

Indicate if you want to offer MICCAI Springer LNCS proceedings to the participants. Publishing a proceedings volume is optional and at the discretion of each challenge's organizers. At a minimum, organizers must ensure that a description of each participant's submission is publicly available. Organizers who wish to publish MICCAI Springer LNCS proceedings must adhere to the MICCAI Satellite events publication process.

Yes

Collaboration with European Society of Radiology (ESR)

In collaboration with European Society of Radiology (ESR), we announce special clinical interest topics with associated clinicians who can help with the preparation of the proposals; the best 3 challenge proposals on these topics will get the opportunity to present their challenges at the European Congress of Radiology (ECR) 2027 in a special session. If you want to organize a challenge in collaboration with ESR on one of these topics, please reach out to the MICCAI Challenges Team (miccai-challenges-2026@dkfz-heidelberg.de) and we will put you in contact with the corresponding clinician.

Challenge in collaboration with ESR. Ticking 'Yes' implies that the challenge has been prepared in collaboration with the clinical contact point.

No

Space and hardware requirements

Organizers of on-site challenges must provide a fair computing environment for all participants. For instance, algorithms should run on the same computing platform provided to all.

We expect the CATHACTION to be an on-site challenge. For this, we would be grateful if we could have access to a large meeting room equipped with computers, projectors, large screen monitors, loudspeakers, and several microphones to accommodate participants and facilitate audience questions.

TASK 1: TASK 1: Catheter and Guidewire Segmentation in Endovascular Intervention

SUMMARY

Abstract

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Endovascular procedures have become a cornerstone of modern cardiovascular care, enabling the minimally invasive treatment of a broad range of vascular pathologies with reduced surgical trauma and faster patient recovery. During these interventions, surgeons navigate flexible catheters and guidewires through the vascular system under X-ray fluoroscopic guidance. Accurate visualization of interventional tools is essential for procedural safety, as poor visibility can increase the risk of vessel wall collisions, perforation, or misplacement of therapeutic devices. However, fluoroscopic imaging provides limited soft-tissue contrast and suffers from noise, motion blur, and occlusion, making catheter and guidewire localization a technically challenging task.

From a computational perspective, catheter and guidewire segmentation represents a fundamental perception problem at the core of surgical navigation and intraoperative assistance. Automated segmentation enables downstream tasks such as tool tracking, collision detection, procedural phase understanding, and safety alerts. Yet, algorithmic development has been constrained by data scarcity, domain variability, and the inherent difficulty of segmenting thin, elongated structures in low-dose X-ray imagery.

The goal of Task 1 is to benchmark and advance state-of-the-art segmentation methods for catheter and guidewire detection using the newly introduced CATHACTION dataset. With over 600,000 frames collected across silicon phantom systems, animal studies, and real human procedures, this dataset provides a unique opportunity to investigate robustness, domain adaptation, and cross-domain generalization. The envisioned biomedical impact includes improved intraoperative awareness and safer endovascular navigation, while the technical impact lies in developing scalable perception models capable of supporting clinical-grade endovascular assistance systems and future foundation models for surgical AI.

Keywords

List the primary keywords that characterize the task.

Catheter and Guidewire Segmentation

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Baoru Huang (University of Liverpool);
 Ruogu Fang (University of Florida);
 Jianing Qiu (Mohamed bin Zayed University of Artificial Intelligence)
 Colin Bicknell (Imperial College London);
 Shuo Li (Case Western Reserve University);
 Francisco Vasconcelos (University College London);
 Danail Stoyanov (University College London);
 Anh Nguyen (University of Liverpool);

b) Provide information on the primary contact person.

Baoru Huang (University of Liverpool),
 Baoru.huang@liverpool.ac.uk

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes, Colin Bicknell (Imperial College London) is a clinician.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

N/A

b) Report the platform used to run the challenge.

We will use [Grand-challenge.org](https://grand-challenge.org) or synapse.org as the platform.

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

Dataset website: <https://airvlab.github.io/cathaction/> - Website for challenge will be updated soon.

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

No user interaction is allowed at any step

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

Participants are allowed to use the data provided by the challenge as well as additional publicly available datasets and pre-trained models. Any external data used must be publicly accessible at the time of the challenge launch. The use of private or proprietary clinical datasets is not permitted. Participants may create additional annotations on publicly available data, provided that these annotations are not based on private data sources.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

There will be cash awards and certificates for the winner and first runner-up in each sub-task provided at least 3 teams submit the results. Cash awards will be subject to the availability of funds from the sponsors. Contacts will be made for taking sponsors onboard.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

Final rankings will be announced publicly during the challenge event and on the official challenge website. The top 3 performing teams for each task will be highlighted. All participating teams will receive their individual performance scores. Detailed evaluation results will be made available after the challenge to ensure transparency and reproducibility.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

All participating teams will submit a brief methodology report in MICCAI format (maximum 4 pages) describing their approach.

The top-performing teams from each sub-task will be invited to contribute to a joint challenge publication. Each

invited team may nominate up to two representatives as co-authors. Authorship will follow standard academic contribution guidelines.

The organizers intend to submit the joint challenge paper within approximately 6–8 months after the challenge. Participating teams are free to publish their methods and results independently. However, teams are requested to appropriately cite the challenge description and dataset paper.

To ensure coordinated dissemination of the official test set results, a short embargo period of up to 3 months after the challenge event will apply to publications that rely primarily on the unpublished test set results. After this period, participants may freely submit and publish their work.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Submissions will be handled through an online evaluation platform (e.g., Grand Challenge or Synapse).

Participants will submit their algorithms as Docker containers that generate predictions on the hidden test set.

Each submission must include:

- a Docker container implementing the inference pipeline
- a result file in the predefined format
- a short description of the method

Detailed submission instructions, including container requirements, data input/output structure, and evaluation workflow, will be provided on the official challenge website prior to the challenge launch.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

Participants will be able to evaluate their algorithms on a validation set prior to the final submission. Multiple submissions will be allowed during the development phase to facilitate method improvement.

A public validation leaderboard will be available to provide performance feedback. However, the final test set will remain hidden, and no performance scores on the test set will be released before the official challenge evaluation.

Each team may submit multiple results, but only the last valid submission before the deadline will be used for the final ranking and official challenge results.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

Challenge website and registration open: 1 April 2026

Training data release: 10 April 2026

Team registration period: 10 July 2026 – 15 August 2026

Validation data release: 10 July 2026

Submission deadline (Docker container, output files, and methodology report): 23 August 2026

Challenge day and spotlight presentations: MICCAI 2026 Workshop (date to be confirmed)

Announcement of challenge winners: during the MICCAI 2026 challenge event

Public release of final rankings and evaluation results: within 2 weeks after the challenge

Joint challenge paper preparation and submission: March 2027

The official challenge website and submission instructions will be available at: [URL to be added]

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

The data for all the tasks come from the CathAction dataset, which is fully anonymised and ethically approved.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)

- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

The evaluation software, including scripts for metric computation and ranking generation, will be made publicly available to ensure transparency and reproducibility of the challenge results.

The code will be released via a public repository (e.g., GitHub) and will include documentation, example inputs, and usage instructions.

The evaluation tools will support standard platforms (Linux-based environments and Docker containers) and will be compatible with the submission format defined for the challenge. The evaluation code will remain available after the challenge to support future benchmarking and reproducibility.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

Participating teams are encouraged, but not required, to make their source code publicly available to promote transparency and reproducibility. Teams choosing to release their code are recommended to do so via public repositories (e.g., GitHub) after the challenge.

Submission of code is not mandatory for participation. However, teams contributing to the joint challenge publication may be invited to share implementation details to support reproducibility.

Any released code should not contain private data or restricted clinical information and should comply with the challenge data usage policies.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

The challenge is organized as an academic initiative without involvement from commercial sponsors that could influence the evaluation process. Any institutional support is limited to infrastructure and data preparation.

Access to the test set labels will be strictly restricted. The ground-truth annotations for the hidden test data will be accessible only to a designated data management team responsible for maintaining the evaluation pipeline. Challenge organizers and participants will not have access to the test labels prior to the final evaluation.

The evaluation will be conducted through an automated pipeline to ensure fairness and reproducibility.

Organizers will not participate in the competition as competing teams and will not submit algorithms for ranking.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Intervention planning, Decision support, Surgery, Training, Assistance

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Localization, Segmentation, Prediction, Tracking

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort consists of adult patients undergoing fluoroscopy-guided endovascular interventions involving catheter and guidewire navigation in major vascular territories. The intended clinical application targets real patients requiring minimally invasive vascular access for diagnostic or therapeutic procedures, where accurate tool localization and collision avoidance are critical for procedural safety.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort consists of fluoroscopic data acquired from three complementary environments: silicon-based vascular phantom systems, preclinical animal studies, and real human clinical procedures. These datasets were collected under realistic interventional workflows and include procedures performed by experienced clinicians. The multi-domain composition enables evaluation of domain generalization across controlled training environments and real-world clinical conditions.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

The challenge is based on X-ray fluoroscopy imaging acquired during endovascular interventions. The dataset includes continuous fluoroscopic video frames collected in silicon-based vascular phantom systems, preclinical animal studies, and real human clinical procedures.

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

In addition to the fluoroscopic image sequences, contextual information related to the imaging data may be provided, including procedural phase indicators, device presence (catheter/guidewire), and frame-level annotations such as segmentation masks and collision labels where applicable.

b) ... to the patient in general (e.g. sex, medical history).

No personally identifiable patient information (e.g., name, date of birth, or hospital identifiers) will be included. Data from phantom and animal studies will not include patient-related attributes.

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The image data originate from fluoroscopy-guided catheterization procedures targeting major vascular structures and related vascular access pathways.

For the target cohort, the data correspond to real patients undergoing minimally invasive endovascular interventions in clinical settings.

For the challenge cohort, the dataset includes fluoroscopic recordings acquired from silicon vascular phantom training systems, preclinical animal experiments, and human clinical procedures, covering a spectrum from controlled environments to real intraoperative scenarios.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The participating algorithms are designed to focus on the detection and segmentation of interventional devices, specifically catheters and guidewires, in fluoroscopic image sequences.

In the target clinical application, the algorithms aim to support real-time perception and safety monitoring during endovascular procedures by identifying device location and potential collisions with vascular structures.

In the challenge setting, the algorithms will operate on multimodal fluoroscopic data from phantom, animal, and human domains to evaluate segmentation performance and domain generalization.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Precision, Accuracy, Reliability

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The imaging data were acquired using clinical X-ray fluoroscopy systems routinely employed in endovascular interventions. The dataset includes recordings from silicon-based vascular phantom training systems, preclinical animal studies, and human clinical procedures.

Standard interventional devices such as catheters and guidewires were used during the procedures. No additional tracking systems were used for performance assessment; instead, manual annotations verified by expert clinicians serve as the reference standard.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Fluoroscopic image sequences were captured during catheterization procedures under routine interventional workflows. Image acquisition followed standard fluoroscopy protocols used in training environments, animal studies, and clinical practice.

Procedures were performed using experienced operators navigating catheters and guidewires through major vascular structures under continuous fluoroscopic guidance. Image resolution, frame rates, and acquisition parameters reflect typical conditions encountered in real endovascular procedures.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

The data were collected across multiple acquisition environments, including clinical institutions and controlled experimental settings used for phantom and preclinical studies.

Specific institution names are not disclosed to preserve data anonymization and comply with data governance regulations. All datasets were obtained following appropriate ethical approvals and data protection protocols.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

The procedures were performed by experienced interventional clinicians trained in fluoroscopy-guided catheterization. Operators included professional surgeons and clinicians with expertise in vascular access and endovascular navigation.

The dataset includes procedures conducted in phantom systems, animal studies, and real clinical environments, reflecting a range of procedural complexities and operator interactions.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case corresponds to one fluoroscopy-guided catheterization procedure. Each case includes a sequence of fluoroscopic image frames capturing catheter and guidewire navigation within vascular structures.

For training and validation cases, image sequences are accompanied by frame-level annotations such as segmentation masks and collision labels. Test cases consist of fluoroscopic image sequences without publicly available annotations, and algorithm outputs are evaluated against hidden ground-truth annotations.

b) State the total number of training, validation and test cases.

There are a total of 650 videos (cases) in our dataset. The dataset comprises fluoroscopic sequences collected from multiple endovascular procedures.

Task 1 (Segmentation): approximately 40,000 annotated frames distributed across multiple procedures.

c) How much of the data are already annotated (stratified by train test in percentage)?

For Task 1 (Segmentation), 100% of the 40,000 frames are annotated with pixel-level segmentation masks of catheter and guidewire.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The dataset split into training (70%), validation (15%), and test (15%) subsets is designed to ensure fair benchmarking and to prevent data leakage across domains.

The training set supports model development across phantom, animal, and human environments, while the validation set enables iterative evaluation. The test set is held out for final performance assessment to ensure unbiased comparison and generalization analysis across acquisition conditions.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The dataset reflects real-world procedural distributions rather than artificially balanced classes.

For segmentation (Task 1), the presence of catheter and guidewire structures varies naturally across frames depending on procedural stage and anatomical region. Challenging scenarios such as vessel bifurcations, occlusions, overlapping anatomical structures, motion blur, and low-contrast conditions are included in all subsets.

The training, validation, and test splits are performed at the procedure (case) level to prevent frame-level data leakage. Domain diversity (phantom, animal, and human data) is preserved across splits to enable evaluation of cross-domain generalization.

This design ensures that performance reflects clinically realistic conditions rather than artificially simplified distributions.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

The dataset used in this challenge consists of newly curated and annotated fluoroscopic sequences collected specifically for the CATHACTION benchmark. A portion of the dataset, particularly the test subset, includes

previously unpublished and unseen data that have not been released publicly.

Approximately 15% of the cases are reserved as a hidden test set and are not accessible to participants. These cases are used exclusively for final evaluation to ensure unbiased benchmarking and to prevent overfitting.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

The reference annotations were generated through manual pixel-level annotation of catheter and guidewire structures in fluoroscopic frames.

For Task 1 (Segmentation), each annotated frame includes detailed segmentation masks distinguishing catheter and guidewire as separate classes.

Annotations were created by a team of approximately 20 trained annotators. To ensure quality and clinical relevance, all annotations were reviewed and verified by two experienced interventional surgeons.

Training and validation annotations are provided to participants, while test annotations are kept hidden and used exclusively for final evaluation.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

Before starting the annotation process, annotators were verbally trained by an expert surgeon to ensure high-quality annotations. The training covered key aspects, including recognizing and differentiating catheters, guidewires, and vascular structures in fluoroscopic images, accurately labeling their boundaries at the pixel level, and maintaining consistency across varying image conditions. Annotators were instructed to resolve ambiguities in low-contrast or occluded regions using contextual clues and flag challenging cases for expert review. Practical guidance on using the annotation tool was also provided, along with self-check protocols to ensure precision before submitting annotations for review by the surgeons.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Annotations were performed by a team of approximately 20 trained annotators with experience in medical image labeling.

All annotations were reviewed and verified by two expert interventional surgeons with extensive clinical experience in fluoroscopy-guided endovascular procedures.

The expert reviewers ensured clinical plausibility, structural consistency, and alignment with procedural workflow, thereby reducing inter-observer variability and improving annotation reliability.

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Each frame was initially annotated by a trained annotator and subsequently reviewed by expert clinicians. In cases where discrepancies were identified, annotations were refined through an expert-guided consensus process.

The final reference annotation for each frame corresponds to the consensus-approved annotation verified by expert interventional surgeons. This process was applied consistently across training, validation, and test cases.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

Before release to participants, fluoroscopic image sequences were standardized through minimal preprocessing steps to ensure consistency while preserving clinically relevant characteristics. These steps include:

anonymization and removal of patient identifiers

frame normalization and consistent image formatting

synchronization of annotation masks with image sequences

No artificial enhancement, filtering, or domain-specific corrections were applied to avoid introducing bias. The same preprocessing pipeline was used for training, validation, and test datasets.

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

Potential sources of annotation error include:

1. low contrast between devices and background structures
2. motion blur and fluoroscopic noise
3. overlapping anatomical structures
4. ambiguity in catheter/guidewire boundaries

Annotation variability may arise from inter-annotator differences and intra-annotator consistency. These uncertainties are expected to be localized primarily around device boundaries and occlusion regions.

To mitigate these effects, annotations were reviewed and verified by expert clinicians, and uncertain cases were resolved through consensus.

b) In an analogous manner, describe and quantify other relevant sources of error.

Additional sources of error may arise from:

1. variation in fluoroscopy acquisition settings across environments
2. procedural variability and operator techniques
3. anatomical differences across subjects

These factors reflect real-world variability and are intentionally preserved to evaluate algorithm robustness and generalization.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

Performance for Task 1 (Segmentation) will be evaluated using the following metrics:

Dice Similarity Coefficient (DSC)

Jaccard Index (IoU)

mean Intersection over Union (mIoU)

Pixel-wise Accuracy

The primary ranking metric will be Dice Similarity Coefficient.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

Dice and IoU are widely used metrics for assessing segmentation performance in medical imaging and are particularly suitable for evaluating overlap between predicted and ground-truth device structures.

In the context of endovascular interventions, accurate delineation of catheter and guidewire boundaries is critical for procedural safety and navigation support. These metrics directly reflect the clinical requirement of precise device localization under fluoroscopic guidance.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

To compute a performance rank for all submitted algorithms, metrics like Dice Score, Jaccard Index, AP, and mAP are first calculated for each test case and submission. These results are then aggregated by averaging the scores for each metric across all test cases. scores, often using a weighted sum or average based on the task's priorities. The algorithms are then ranked in descending order of their final scores, ensuring that all aspects of performance, such as segmentation accuracy and collision detection precision, are comprehensively evaluated to produce a fair ranking.

b) Describe the method(s) used to manage submissions with missing results on test cases.

Submissions that fail to produce results for specific test cases will be assigned the lowest possible score for those cases.

If a submission fails on more than a predefined proportion of cases, it may be excluded from the final ranking to ensure fairness and comparability across participating algorithms.

c) Justify why the described ranking scheme(s) was/were used.

The ranking is based on the evaluation metric score obtained on the private test set. This approach ensures that algorithms are ranked according to their true performance on unseen data, maintaining fairness and accuracy in the evaluation process.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

Statistical analysis will be conducted to assess segmentation performance, robustness, and variability across cases and algorithms. Performance will be evaluated at the case level using standard segmentation metrics, and aggregated to obtain overall estimates.

Confidence intervals, variability measures, and statistical significance testing will be applied to ensure that performance differences between algorithms are interpretable and not due to random variation. These statistical approaches are commonly used in medical image analysis challenges and support reliable benchmarking under heterogeneous clinical conditions.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of performance estimates for each algorithm will be assessed using confidence intervals computed on the test set.

Specifically, 95% confidence intervals for the mean Dice Similarity Coefficient will be estimated using percentile bootstrap resampling at the case level. This approach accounts for variability across procedures and provides robust uncertainty estimates without assuming normality.

Provide a description of how variability of the performance of individual algorithms across test cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

Variability of algorithm performance across test cases will be assessed using descriptive statistics including standard deviation (SD), interquartile range (IQR), and distribution plots of Dice scores across cases.

Outlier cases with unusually low or high performance will be identified and analyzed to understand algorithm robustness under challenging procedural conditions.

Provide a description of how variability of rankings is assessed.

Ranking stability will be assessed using bootstrap resampling of test cases. For each resampling iteration, algorithms will be re-ranked based on mean Dice performance.

The variability of rankings will be summarized using rank distributions and confidence intervals, providing insight into whether the relative ordering of algorithms is robust or sensitive to case composition.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

Pairwise statistical comparisons between top-performing algorithms will be conducted using non-parametric tests (e.g., Wilcoxon signed-rank test) on case-level Dice scores.

This approach evaluates whether observed performance differences are statistically significant without assuming normal distribution of scores. Multiple comparison corrections will be applied where appropriate.

Provide a description of the missing data handling.

Missing predictions for specific test cases will be treated as failed cases and assigned the lowest possible performance score for those cases.

Submissions with missing outputs exceeding a predefined threshold may be excluded from the final ranking to ensure fairness and comparability.

Indicate any software product that is used for all data analysis methods.

Statistical analyses will be performed using standard scientific computing environments, including Python-based libraries (e.g., NumPy, SciPy, scikit-learn) and R where appropriate.

Evaluation scripts will be made available to ensure reproducibility of the statistical analysis.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

Additional analyses will be conducted to better understand algorithm behavior and benchmarking outcomes, including:

1. performance comparison across domains (phantom, animal, human)
2. inter-algorithm variability and robustness analysis
3. identification of common failure modes and challenging cases
4. exploratory analysis of potential performance improvements through ensemble strategies

These analyses aim to provide deeper insights into algorithm generalization and clinical applicability beyond simple leaderboard rankings.

TASK 2: TASK 2: Collision Detection in Endovascular Intervention

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

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Endovascular interventions are performed in highly delicate vascular territories, such as the coronary and cerebral circulations, where even minor mechanical trauma can have catastrophic consequences. During these minimally invasive procedures, flexible catheters and guidewires are steered through tortuous vasculature under X-ray fluoroscopic guidance. Unintended collisions between instruments and vessel walls, or with calcified plaques and aneurysm sacs, can cause dissection, perforation, thromboembolic events, or hemorrhage. As endovascular workflows become more complex and increasingly incorporate robotic assistance and AI-based guidance, reliable, real-time collision detection is emerging as a central requirement for ensuring patient safety and enabling truly intelligent support systems.

From a technical standpoint, collision detection in endovascular intervention is an intrinsically challenging problem. The vascular environment is constrained, deformable, and dynamic; vessels vary in diameter, curvature, and compliance, and their appearance changes with contrast injection and motion. Fluoroscopic imaging provides only noisy 2D projections of a 3D scene, with overlapping anatomy and low contrast for thin instruments. Furthermore, the highly flexible catheters and guidewires exhibit non-linear interactions with surrounding tissues, making it difficult to infer contact or collision states from image data alone. Addressing this problem requires models that jointly reason about instrument shape, proximity to vessel walls, and temporal evolution of the procedure.

Task 2 of the CATHACTION challenge is designed to catalyze progress on this safety-critical problem by benchmarking methods for collision detection using a large-scale, multi-domain dataset of fluoroscopic sequences. Leveraging over 600+ videos from silicon phantom systems, animal X-ray studies, and real human procedures, the task explicitly encourages approaches that are robust across domains and suitable for real-time deployment. The envisioned biomedical impact is improved intraoperative safety, reduced vascular complications, and better support for both novice and expert operators, including in robotic-assisted settings. The technical impact includes advancing spatio-temporal modeling, multi-task learning in combination with tool segmentation, and domain-adaptive algorithms that move surgical AI closer to reliable, clinically deployable endovascular assistance.

Keywords

List the primary keywords that characterize the task.

Collision Detection

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Baoru Huang (University of Liverpool);

Ruogu Fang (University of Florida);

Jianing Qiu (Mohamed bin Zayed University of Artificial Intelligence)

Colin Bicknell (Imperial College London);

Shuo Li (Case Western Reserve University);

Francisco Vasconcelos (University College London);

Danail Stoyanov (University College London);

Anh Nguyen (University of Liverpool);

b) Provide information on the primary contact person.

Baoru Huang (University of Liverpool),

Baoru.huang@liverpool.ac.uk

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes, Colin Bicknell (Imperial College London) is a clinician.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

N/A

b) Report the platform used to run the challenge.

We will use [Grand-challenge.org](https://grand-challenge.org) or synapse.org as the platform.

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

Dataset website: <https://airvlab.github.io/cathaction/> - Website for challenge will be updated soon.

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

No user interaction is allowed at any step

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

Participants are allowed to use the data provided by the challenge as well as additional publicly available datasets and pre-trained models. Any external data used must be publicly accessible at the time of the challenge launch. The use of private or proprietary clinical datasets is not permitted. Participants may create additional annotations on publicly available data, provided that these annotations are not based on private data sources.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

There will be cash awards and certificates for the winner and first runner-up in each sub-task provided at least 3 teams submit the results. Cash awards will be subject to the availability of funds from the sponsors. Contacts will be made for taking sponsors onboard.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

Final rankings will be announced publicly during the challenge event and on the official challenge website. The top 3 performing teams for each task will be highlighted. All participating teams will receive their individual performance scores. Detailed evaluation results will be made available after the challenge to ensure transparency and reproducibility.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

All participating teams will submit a brief methodology report in MICCAI format (maximum 4 pages) describing their approach.

The top-performing teams from each sub-task will be invited to contribute to a joint challenge publication. Each invited team may nominate up to two representatives as co-authors. Authorship will follow standard academic contribution guidelines.

The organizers intend to submit the joint challenge paper within approximately 6–8 months after the challenge. Participating teams are free to publish their methods and results independently. However, teams are requested to appropriately cite the challenge description and dataset paper.

To ensure coordinated dissemination of the official test set results, a short embargo period of up to 3 months after the challenge event will apply to publications that rely primarily on the unpublished test set results. After this period, participants may freely submit and publish their work.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Submissions will be handled through an online evaluation platform (e.g., Grand Challenge or Synapse).

Participants will submit their algorithms as Docker containers that generate predictions on the hidden test set.

Each submission must include:

- a Docker container implementing the inference pipeline
- a result file in the predefined format
- a short description of the method

Detailed submission instructions, including container requirements, data input/output structure, and evaluation workflow, will be provided on the official challenge website prior to the challenge launch.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

Participants will be able to evaluate their algorithms on a validation set prior to the final submission. Multiple submissions will be allowed during the development phase to facilitate method improvement.

A public validation leaderboard will be available to provide performance feedback. However, the final test set will remain hidden, and no performance scores on the test set will be released before the official challenge

evaluation.

Each team may submit multiple results, but only the last valid submission before the deadline will be used for the final ranking and official challenge results.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

Challenge website and registration open: 1 April 2026

Training data release: 10 April 2026

Team registration period: 10 July 2026 – 15 August 2026

Validation data release: 10 July 2026

Submission deadline (Docker container, output files, and methodology report): 23 August 2026

Challenge day and spotlight presentations: MICCAI 2026 Workshop (date to be confirmed)

Announcement of challenge winners: during the MICCAI 2026 challenge event

Public release of final rankings and evaluation results: within 2 weeks after the challenge

Joint challenge paper preparation and submission: March 2027

The official challenge website and submission instructions will be available at: [URL to be added]

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

The data for all the tasks come from the CathAction dataset, which is fully anonymised and ethically approved.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

The evaluation software, including scripts for metric computation and ranking generation, will be made publicly available to ensure transparency and reproducibility of the challenge results.

The code will be released via a public repository (e.g., GitHub) and will include documentation, example inputs, and usage instructions.

The evaluation tools will support standard platforms (Linux-based environments and Docker containers) and will be compatible with the submission format defined for the challenge. The evaluation code will remain available after the challenge to support future benchmarking and reproducibility.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

Participating teams are encouraged, but not required, to make their source code publicly available to promote transparency and reproducibility. Teams choosing to release their code are recommended to do so via public repositories (e.g., GitHub) after the challenge.

Submission of code is not mandatory for participation. However, teams contributing to the joint challenge publication may be invited to share implementation details to support reproducibility.

Any released code should not contain private data or restricted clinical information and should comply with the challenge data usage policies.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

The challenge is organized as an academic initiative without involvement from commercial sponsors that could influence the evaluation process. Any institutional support is limited to infrastructure and data preparation.

Access to the test set labels will be strictly restricted. The ground-truth annotations for the hidden test data will be

accessible only to a designated data management team responsible for maintaining the evaluation pipeline. Challenge organizers and participants will not have access to the test labels prior to the final evaluation.

The evaluation will be conducted through an automated pipeline to ensure fairness and reproducibility. Organizers will not participate in the competition as competing teams and will not submit algorithms for ranking.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Assistance, Decision support, Intervention planning, Training, Surgery

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation

- Tracking

Prediction, Detection, Tracking, Localization

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort consists of adult patients undergoing fluoroscopy-guided endovascular interventions involving catheter and guidewire navigation in major vascular territories. The intended clinical application targets real patients requiring minimally invasive vascular access for diagnostic or therapeutic procedures, where accurate tool localization and collision avoidance are critical for procedural safety.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort consists of fluoroscopic data acquired from three complementary environments: silicon-based vascular phantom systems, preclinical animal studies, and real human clinical procedures. These datasets were collected under realistic interventional workflows and include procedures performed by experienced clinicians. The multi-domain composition enables evaluation of domain generalization across controlled training environments and real-world clinical conditions.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

The challenge is based on X-ray fluoroscopy imaging acquired during endovascular interventions. The dataset includes continuous fluoroscopic video frames collected in silicon-based vascular phantom systems, preclinical animal studies, and real human clinical procedures.

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

In addition to the fluoroscopic image sequences, contextual information related to the imaging data may be provided, including procedural phase indicators, device presence (catheter/guidewire), and frame-level annotations such as segmentation masks and collision labels where applicable.

b) ... to the patient in general (e.g. sex, medical history).

No personally identifiable patient information (e.g., name, date of birth, or hospital identifiers) will be included. Data from phantom and animal studies will not include patient-related attributes.

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The image data originate from fluoroscopy-guided catheterization procedures targeting major vascular structures and related vascular access pathways.

For the target cohort, the data correspond to real patients undergoing minimally invasive endovascular interventions in clinical settings.

For the challenge cohort, the dataset includes fluoroscopic recordings acquired from silicon vascular phantom training systems, preclinical animal experiments, and human clinical procedures, covering a spectrum from controlled environments to real intraoperative scenarios.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The participating algorithms are designed to focus on the detection and segmentation of interventional devices, specifically catheters and guidewires, in fluoroscopic image sequences.

In the target clinical application, the algorithms aim to support real-time perception and safety monitoring during endovascular procedures by identifying device location and potential collisions with vascular structures.

In the challenge setting, the algorithms will operate on multimodal fluoroscopic data from phantom, animal, and human domains to evaluate segmentation performance and domain generalization.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Precision, Accuracy, Reliability

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The imaging data were acquired using clinical X-ray fluoroscopy systems routinely employed in endovascular interventions. The dataset includes recordings from silicon-based vascular phantom training systems, preclinical animal studies, and human clinical procedures.

Standard interventional devices such as catheters and guidewires were used during the procedures. No additional tracking systems were used for performance assessment; instead, manual annotations verified by expert clinicians serve as the reference standard.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Fluoroscopic image sequences were captured during catheterization procedures under routine interventional workflows. Image acquisition followed standard fluoroscopy protocols used in training environments, animal studies, and clinical practice.

Procedures were performed using experienced operators navigating catheters and guidewires through major vascular structures under continuous fluoroscopic guidance. Image resolution, frame rates, and acquisition parameters reflect typical conditions encountered in real endovascular procedures.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

The data were collected across multiple acquisition environments, including clinical institutions and controlled experimental settings used for phantom and preclinical studies.

Specific institution names are not disclosed to preserve data anonymization and comply with data governance regulations. All datasets were obtained following appropriate ethical approvals and data protection protocols.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

The procedures were performed by experienced interventional clinicians trained in fluoroscopy-guided catheterization. Operators included professional surgeons and clinicians with expertise in vascular access and endovascular navigation.

The dataset includes procedures conducted in phantom systems, animal studies, and real clinical environments, reflecting a range of procedural complexities and operator interactions.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).

- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case corresponds to one fluoroscopy-guided catheterization procedure. Each case includes a sequence of fluoroscopic image frames capturing catheter and guidewire navigation within vascular structures.

For training and validation cases, image sequences are accompanied by frame-level annotations such as segmentation masks and collision labels. Test cases consist of fluoroscopic image sequences without publicly available annotations, and algorithm outputs are evaluated against hidden ground-truth annotations.

b) State the total number of training, validation and test cases.

There are a total of 650 videos (cases) in our dataset. The dataset comprises fluoroscopic sequences collected from multiple endovascular procedures.

Task 2 (Collision Detection): approximately 600,000 annotated frames across the full dataset.

The dataset is divided into training, validation, and test subsets at the case (procedure) level to prevent data leakage.

c) How much of the data are already annotated (stratified by train test in percentage)?

All frames included in both tasks are fully annotated.

For Task 2 (Collision Detection), 100% of the 600,000 frames are annotated with frame-level collision labels.

Annotations were generated by trained annotators and verified by expert clinicians to ensure quality and consistency.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The dataset split into training (70%), validation (15%), and test (15%) subsets is designed to ensure fair benchmarking and to prevent data leakage across domains.

The training set supports model development across phantom, animal, and human environments, while the validation set enables iterative evaluation. The test set is held out for final performance assessment to ensure unbiased comparison and generalization analysis across acquisition conditions.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The dataset reflects real-world procedural distributions rather than artificially balanced classes.

For segmentation (Task 1), the presence of catheter and guidewire structures varies naturally across frames depending on procedural stage and anatomical region. Challenging scenarios such as vessel bifurcations, occlusions, overlapping anatomical structures, motion blur, and low-contrast conditions are included in all subsets.

The training, validation, and test splits are performed at the procedure (case) level to prevent frame-level data leakage. Domain diversity (phantom, animal, and human data) is preserved across splits to enable evaluation of cross-domain generalization.

This design ensures that performance reflects clinically realistic conditions rather than artificially simplified distributions.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

The dataset used in this challenge consists of newly curated and annotated fluoroscopic sequences collected specifically for the CATHACTION benchmark. A portion of the dataset, particularly the test subset, includes previously unpublished and unseen data that have not been released publicly.

Approximately 15% of the cases are reserved as a hidden test set and are not accessible to participants. These cases are used exclusively for final evaluation to ensure unbiased benchmarking and to prevent overfitting.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

The reference annotations were generated through manual pixel-level annotation of catheter and guidewire structures in fluoroscopic frames.

For Task 1 (Segmentation), each annotated frame includes detailed segmentation masks distinguishing catheter and guidewire as separate classes.

Annotations were created by a team of approximately 20 trained annotators. To ensure quality and clinical relevance, all annotations were reviewed and verified by two experienced interventional surgeons.

Training and validation annotations are provided to participants, while test annotations are kept hidden and used exclusively for final evaluation.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

Before starting the annotation process, annotators were verbally trained by an expert surgeon to ensure high-quality annotations. The training covered key aspects, including recognizing and differentiating catheters, guidewires, and vascular structures in fluoroscopic images, accurately labeling their boundaries at the pixel level, and maintaining consistency across varying image conditions. Annotators were instructed to resolve ambiguities in low-contrast or occluded regions using contextual clues and flag challenging cases for expert review. Practical guidance on using the annotation tool was also provided, along with self-check protocols to ensure precision before submitting annotations for review by the surgeons.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Annotations were performed by a team of approximately 20 trained annotators with experience in medical image labeling.

All annotations were reviewed and verified by two expert interventional surgeons with extensive clinical experience in fluoroscopy-guided endovascular procedures.

The expert reviewers ensured clinical plausibility, structural consistency, and alignment with procedural workflow, thereby reducing inter-observer variability and improving annotation reliability.

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Each frame was initially annotated by a trained annotator and subsequently reviewed by expert clinicians. In cases where discrepancies were identified, annotations were refined through an expert-guided consensus process.

The final reference annotation for each frame corresponds to the consensus-approved annotation verified by expert interventional surgeons. This process was applied consistently across training, validation, and test cases.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

Before release to participants, fluoroscopic image sequences were standardized through minimal preprocessing steps to ensure consistency while preserving clinically relevant characteristics. These steps include:

anonymization and removal of patient identifiers

frame normalization and consistent image formatting

synchronization of annotation masks with image sequences

No artificial enhancement, filtering, or domain-specific corrections were applied to avoid introducing bias. The same preprocessing pipeline was used for training, validation, and test datasets.

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

Potential sources of annotation error include:

1. low contrast between devices and background structures

2. motion blur and fluoroscopic noise
3. overlapping anatomical structures
4. ambiguity in catheter/guidewire boundaries

Annotation variability may arise from inter-annotator differences and intra-annotator consistency. These uncertainties are expected to be localized primarily around device boundaries and occlusion regions.

To mitigate these effects, annotations were reviewed and verified by expert clinicians, and uncertain cases were resolved through consensus.

b) In an analogous manner, describe and quantify other relevant sources of error.

Additional sources of error may arise from:

1. variation in fluoroscopy acquisition settings across environments
2. procedural variability and operator techniques
3. anatomical differences across subjects

These factors reflect real-world variability and are intentionally preserved to evaluate algorithm robustness and generalization.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

Performance of the submitted algorithms for collision detection will be evaluated using Average Precision (AP) and mean Average Precision (mAP).

AP is computed based on precision–recall curves for collision event detection under different confidence thresholds. mAP is calculated by averaging AP scores across all test cases and across defined detection settings.

The primary ranking metric will be mAP on the private test set, while AP will be reported as a complementary performance indicator.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

AP and mAP are widely adopted evaluation metrics in object detection and event detection tasks, and are particularly suitable for collision detection in fluoroscopic sequences.

Collision detection requires both accurate localization and reliable discrimination between true collision and non-collision events. AP captures the balance between precision and recall under different thresholds, ensuring

that algorithms are penalized for both missed collisions and false alarms. mAP provides a robust overall performance estimate across varying procedural scenarios and imaging conditions.

These metrics reflect clinically relevant performance requirements, where both sensitivity (detecting potential hazards) and specificity (avoiding unnecessary alerts) are essential for intraoperative safety support.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

For each submission, AP is computed per test case based on predicted collision events and reference annotations.

The ranking procedure follows three steps:

Compute AP for each test case

Aggregate results across the entire test set to obtain mean AP (mAP)

Rank all submissions in descending order based on mAP. The final leaderboard will be determined using mAP as the primary ranking metric.

b) Describe the method(s) used to manage submissions with missing results on test cases.

Submissions that fail to produce results for specific test cases will be assigned the lowest possible score for those cases.

If a submission fails on more than a predefined proportion of cases, it may be excluded from the final ranking to ensure fairness and comparability across participating algorithms.

c) Justify why the described ranking scheme(s) was/were used.

The ranking is based on mean Average Precision (mAP) across all test cases to ensure a consistent and interpretable evaluation of collision detection performance in clinically realistic settings.

This strategy:

1. reflects clinically meaningful performance across diverse procedural scenarios
2. avoids bias toward individual cases or specific imaging conditions
3. ensures fair comparison between algorithms
4. aligns with established evaluation practices in medical image detection challenges

Using mAP as the primary ranking metric provides a robust assessment of an algorithm's ability to detect potential hazardous interactions between instruments and vascular structures, which is critical for intraoperative safety monitoring and decision support.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

Statistical analysis will be conducted to assess segmentation performance, robustness, and variability across cases and algorithms. Performance will be evaluated at the case level using standard segmentation metrics, and aggregated to obtain overall estimates.

Confidence intervals, variability measures, and statistical significance testing will be applied to ensure that performance differences between algorithms are interpretable and not due to random variation. These statistical approaches are commonly used in medical image analysis challenges and support reliable benchmarking under heterogeneous clinical conditions.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of performance estimates for each algorithm will be assessed using confidence intervals computed on the test set.

Specifically, 95% confidence intervals for the mean Dice Similarity Coefficient will be estimated using percentile bootstrap resampling at the case level. This approach accounts for variability across procedures and provides robust uncertainty estimates without assuming normality.

Provide a description of how variability of the performance of individual algorithms across test cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

Variability of algorithm performance across test cases will be assessed using descriptive statistics including standard deviation (SD), interquartile range (IQR), and distribution plots of Dice scores across cases.

Outlier cases with unusually low or high performance will be identified and analyzed to understand algorithm robustness under challenging procedural conditions.

Provide a description of how variability of rankings is assessed.

Ranking stability will be assessed using bootstrap resampling of test cases. For each resampling iteration, algorithms will be re-ranked based on mean Dice performance.

The variability of rankings will be summarized using rank distributions and confidence intervals, providing insight into whether the relative ordering of algorithms is robust or sensitive to case composition.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

Pairwise statistical comparisons between top-performing algorithms will be conducted using non-parametric tests (e.g., Wilcoxon signed-rank test) on case-level Dice scores.

This approach evaluates whether observed performance differences are statistically significant without assuming normal distribution of scores. Multiple comparison corrections will be applied where appropriate.

Provide a description of the missing data handling.

Missing predictions for specific test cases will be treated as failed cases and assigned the lowest possible performance score for those cases.

Submissions with missing outputs exceeding a predefined threshold may be excluded from the final ranking to ensure fairness and comparability.

Indicate any software product that is used for all data analysis methods.

Statistical analyses will be performed using standard scientific computing environments, including Python-based libraries (e.g., NumPy, SciPy, scikit-learn) and R where appropriate.

Evaluation scripts will be made available to ensure reproducibility of the statistical analysis.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

Additional analyses will be conducted to better understand algorithm behavior and benchmarking outcomes, including:

1. performance comparison across domains (phantom, animal, human)
2. inter-algorithm variability and robustness analysis
3. identification of common failure modes and challenging cases
4. exploratory analysis of potential performance improvements through ensemble strategies

These analyses aim to provide deeper insights into algorithm generalization and clinical applicability beyond simple leaderboard rankings.

ADDITIONAL POINTS

References

Please include any reference important for the challenge design, for example publications on the data, the annotation process or the chosen metrics as well as DOIs referring to data or code.

Preprint paper: <https://arxiv.org/abs/2408.13126>

Dataset website: <https://airvlab.github.io/cathaction>

Further comments

Further comments from the organizers.

N/A